

Scalable Automatic Identification for Hierarchical Causal Models: A New Look at Project STAR

Anass Ettahiri,¹ Yasser Oufqir,¹ Simon Patry,¹ Aymen El Ouadrhiri,¹ Aghiles Drali,¹ David Cortes,²
Janis Aiad,¹ Emilie Devijver^{3,*} and Marianne Clausel⁴

¹Ecole Polytechnique, France

²Al-vidence, France

³CNRS, Laboratoire d'Informatique de Grenoble, Université Grenoble Alpes, France

⁴CRAN, Simul Research Group, Université de Lorraine, France

*Corresponding author. emilie.devijver@univ-grenoble-alpes.fr

Abstract

Hierarchical data arise when observations are nested within larger units, as in students within classes, cells within patients, or citizens within states. Although structural causal models and the do-calculus provide a mature language for causal identification, they are usually applied to flat data and do not directly encode interventions acting on distributions of sub-units within units. We develop a scalable automatic identification and estimation implementation for Hierarchical Causal Models (HCM), combining graph transformations, pyAgrum do-calculus identification, adaptation of the returned symbolic expression into closed-form HCM formulas, plug-in density estimation, and numerical evaluation. The adapted abstract syntax tree (AST) is central because it turns pyAgrum's identified formula into local density, expectation, and marginalization tasks that can be fitted or evaluated independently. The implementation and STAR replication scripts are available at this link. The application to Project STAR motivates the paper: class-size assignment is randomized at the class level, while outcomes and covariates are observed at the student level, so the natural intervention is a class-level intervention on the distribution of treatment, not only a pooled student-level indicator. We validate the pipeline on canonical HCM motifs and benchmark scenarios with known ground truth, then analyse STAR kindergarten mathematics outcomes. OLS, instrumental-variable, and non-causal hierarchical Bayesian/OLS baselines give useful reference contrasts, whereas HCM estimates based on discovered graphs are larger and reveal sensitivity to class heterogeneity, graph structure, and gender-dependent multi-parent plug-in factors. The results show that symbolic identification alone is not enough for practical hierarchical causal inference: scalable estimation, diagnostics, and effective graphs are central parts of the scientific object. **mettre en valeur l'utilisation des modèles hiérarchiques plutôt que à plat**

Keywords hierarchical causal models, automatic identification, scalable estimation, do-calculus, Project STAR, class-size effects, plug-in estimation

Key Messages

We provide software for automatic HCM identification and scalable plug-in estimation. Applied to Project STAR, the pipeline turns class heterogeneity into a causal estimand and gives a new interpretation of class-size effects beyond non-causal hierarchical Bayesian/OLS baselines.

Introduction

Many datasets in social, biomedical, and natural sciences are hierarchical. The observational units used for modelling are rarely exchangeable singletons: students are nested in classes and schools, cells are nested in patients, plots are nested in regions, and repeated measures are nested within subjects. These structures are not merely a nuisance for variance estimation. They affect what an intervention means. For instance, a class-size intervention in education is assigned at the class level but outcomes are measured for students. The object of intervention is therefore a distribution of sub-unit treatments inside a unit, not only a binary treatment attached independently to each row of a flat table.

Classical causal inference provides a mature formal language for interventions. Structural causal models (SCMs), causal graphical models, adjustment criteria, instrumental variables, and Pearl's do-calculus identify when a causal estimand can be written as a functional of observed data (Pearl, 2009; Imbens and Rubin, 2015). However, most implementations assume a flat set of random variables. If the analyst aggregates sub-units into class means, information about within-unit heterogeneity is lost. If the analyst instead pools all sub-units, unit-level latent variables induce dependence and confounding that is invisible to the flat model. This tension is exactly what Hierarchical Causal Models (HCM) were designed to address (Weinstein and Blei, 2023). There is also a computational problem. Once an HCM estimand has been

identified, it is usually not a single regression but a product of many unit-level and class-level density factors, conditional response surfaces, and marginalizations over Q -variables. Without a structured representation, estimation becomes a monolithic formula that is hard to audit and hard to scale. Our implementation treats the identified formula as a computational graph, so that many conditional densities, per-unit response models, and Monte Carlo branches can be estimated independently within the same causal framework.

The central device in HCM is the Q -variable: a random distribution at the unit level that represents how a sub-unit variable behaves conditional on its sub-unit parents. In the large-within-unit limit, the hierarchical model can be transformed into a flat causal graphical model over Q -variables and unit-level variables. Identification can then be performed with standard do-calculus on this transformed graph. This paper focuses on the next operational step: turning such formulas into numerical estimators, diagnostics, and real-data workflows.

Our motivating application is Project STAR, the Tennessee Student/Teacher Achievement Ratio experiment (Krueger, 1999). STAR randomized students within schools to small classes, regular classes, or regular classes with an aide. The canonical econometric analysis estimates student-level regressions with school fixed effects. Yet the treatment mechanism is inherently hierarchical: class type is shared by students in the same class, and class composition (gender, socioeconomic status, peer environment) can influence both outcomes and the interpretation of the treatment effect. This application motivates our methodological question: can we implement HCM in a way that carries the analyst from raw nested data to an estimated causal effect, while making explicit which pieces of the identified formula were actually estimated?

Contributions

The contribution of the paper is to make HCM operational as an automatic and scalable statistical workflow. Starting from a hierarchical graph, the implementation collapses the graph, augments it with outcome-distribution nodes when needed, marginalizes irrelevant variables, calls pyAgrum for symbolic do-calculus identification, adapts the returned expression into closed-form HCM formulas, and then evaluates the resulting functional with plug-in density estimators. The identified expression is not treated as a final display object. It is converted into an abstract syntax tree (AST) that simultaneously records the formal estimand, drives numerical evaluation, and exposes which density factors were actually fitted. This adapted AST representation also gives the framework its scalable estimation path: each conditional density node, expectation node, and summation branch is an explicit unit of work, so the same identified formula can be evaluated by fitting many small estimators rather than one opaque global model. This design is validated on canonical motifs, do-calculus galleries, and benchmark scenarios with ground-truth effects, before being used on Project STAR to compare hierarchical causal estimands with OLS, instrumental-variable, and non-causal hierarchical baselines. The code is released as an open repository, [AI-evidence/hierarchicalcausalmodels](#), including the graph transformations, do-calculus interface, AST evaluator, estimators, and STAR scripts used to produce the stable artefacts reported here.

Related work and positioning

The paper builds on the SCM and graphical-model tradition (Pearl, 2009), the potential-outcome/econometric literature on treatment effects (Imbens and Rubin, 2015; Angrist et al., 1996), and the original HCM formalism of Weinstein and Blei (2023). The STAR application connects to the education-production literature initiated by Krueger (1999) and later work on long-run impacts of class assignment (Krueger and Whitmore, 2001; Chetty et al., 2011). The implementation also intersects with causal software: pyAgrum provides graph and do-calculus routines (Gonzales et al., 2017), while modern Python causal libraries increasingly expose symbolic estimands.

The `y0` library recently added symbolic HCM support through a domain-specific language that extends to counterfactual transport, including the HCM pull request in January 2025. Our contribution is different: we implement the path from nested data to estimated ATE, including the adaptation of pyAgrum’s symbolic output into closed-form HCM formulas, density fitting, scalable AST evaluation, diagnostics, and a real STAR workflow rather than formula derivation alone.

The rest of the paper follows this computational path. Section 2 describes the symbolic-to-numeric backend, Section 3 validates the implementation on controlled settings, Section 4 gives the Project STAR analysis, and Section 5 discusses limitations and future work.

Method

Symbolic-to-numeric backend

The implementation is organized as a symbolic–numeric backend rather than as a collection of graph-specific estimators. The analyst provides a hierarchical graph whose variables are typed by level, together with nested arrays whose rows are units and whose columns are sub-units. From that input, the backend follows a single path from the graphical specification to a numerical contrast:

HSCM graph $\longrightarrow$ collapsed/augmented graph $\longrightarrow$ pyAgrum causal model $\longrightarrow$ pyAgrum expression $\longrightarrow$ adapted AST $\longrightarrow$ plug-in estimator.

This separation is central to the implementation. pyAgrum is used only after the HCM graph has been converted into an ordinary causal graph over unit-level quantities, and it returns an identifiable symbolic functional rather than an estimate. The adaptation layer then turns this pyAgrum output into the HCM closed form used in the paper and into an AST whose factors can be represented from data before the final intervention contrast is evaluated. The same separation is what makes the implementation scalable. The symbolic layer decides what has to be estimated; the numerical layer sees a list of conditional density, expectation, and marginalization tasks that can be fitted independently across units, factors, or Monte Carlo samples. Thus parallel estimation is not an external engineering add-on, but a direct consequence of representing the identified estimand as a typed computational graph.

Let $i = 1, \dots, n$ index units and $j = 1, \dots, m_i$ index sub-units. For each sub-unit variable L_{ij} with sub-unit parents $\text{pa}_S(L)_{ij}$, the backend creates a unit-level Q -node

$$Q_i^L | \text{pa}_S(L) = p(L_{ij} | \text{pa}_S(L)_{ij}, \text{unit} = i). \quad (1)$$

Operationally, this equation is treated as a rule for constructing a new node name and its parents. For example, an outcome mechanism $M \leftarrow (A, E, G, L)$ becomes the node $Q^m | a, e, g, l$. The backend stores this node as a variable of the collapsed graph, but its numerical value is only fitted later from the sub-unit table. Thus equation (1) has two roles: graphically, it replaces a sub-unit mechanism by a unit-level variable; numerically, it declares which conditional density estimator will be needed.

The graph passed to pyAgrum is obtained through the HCM transformations. Collapse replaces every sub-unit mechanism by its Q -node and lifts all edges to a flat directed acyclic graph over unit-level variables and Q -variables, denoted G_{col} . Augmentation then adds auxiliary Q -nodes for the target outcome, so that the outcome distribution is explicit in the collapsed graph; when required, marginalization projects out variables that are not part of the identification problem. Only after these transformations is the graph copied into a pyAgrum DAG with the same nodes and directed arcs. The treatment and outcome names are then passed to pyAgrum's do-calculus identification routine. This means that pyAgrum never sees the original nested table: it sees the collapsed HCM graph and returns a symbolic expression for $p(Y | \text{do}(X = x))$ in terms of observed variables of that graph. Keeping this step separate from estimation is what allows the same identified expression to be evaluated later with Gaussian, mixture, categorical, non-parametric, or neural plug-in components.

The symbolic expression returned by pyAgrum is adapted into a typed AST and then written as the closed-form do-calculus formula used by the estimator. Here AST means an abstract syntax tree: a nested object representation of the pyAgrum-identified formula after translation into HCM density factors, not only a string displayed in the paper. Each node has a computational role. A summation node stores the variables to marginalize; a product node stores the multiplicative factors; a conditional node stores one density key of the form $P(\text{outcome} | \text{parents})$; and a leaf node is used for constants or fallback terms. The do-calculus engine demo prints this adapted object directly. For instance, one retained collapsed example identifies

$$\sum_{Q^z | a, W} P(W | Q^a) P(Y | Q^a, Q^z | a, W) \left[\sum_{Q^a} P(Q^z | a | Q^a, W) P(Q^a) \right].$$

The corresponding AST can be read horizontally as

$$\text{Sum}(Q^z | a, W) \implies \text{Product} \implies \frac{\text{Product}\{P(W | Q^a), P(Y | Q^a, Q^z | a, W)\}}{\text{Sum}(Q^a)\{\text{Product}[P(Q^z | a | Q^a, W), P(Q^a)]\}}.$$

This is the object used by the evaluator. The first line is the outer marginalization node over $Q^z | a$ and W . The product node then has two children: a direct product of fitted conditionals, and an inner summation over Q^a . Calling `collect_conditionals()` on this tree returns the density keys $P(W | Q^a)$, $P(Y | Q^a, Q^z | a, W)$, $P(Q^z | a | Q^a, W)$, and $P(Q^a)$ that must be fitted or read from data. Changing from $\text{do}(Q^a = 1)$ to $\text{do}(Q^a = 0)$ does not change the tree structure; it changes only the intervention value used when the same parsed tree is evaluated. In larger formulas, this same mechanism may produce dozens of density keys and many repeated unit-level prediction problems. Those tasks share the same formal estimand but do not need to be fitted as a single global regression, which is why the framework can scale to large nested datasets without changing the causal definition of the target.

Evaluation and diagnostics

The general target is

$$\tau(q_1, q_0) = \mathbb{E}[Y | \text{do}(Q^a = q_1)] - \mathbb{E}[Y | \text{do}(Q^a = q_0)]. \quad (2)$$

In the implementation, equation (2) is evaluated by copying the identified AST twice. The first copy fixes the intervention node at $Q^a = q_1$, the second fixes it at $Q^a = q_0$, and the ATE is the difference between the two numerical evaluations. For a backdoor-like confounder motif, this procedure reduces to the familiar unit-averaged plug-in estimator

$$\hat{\tau}(q_1, q_0) = \frac{1}{n} \sum_{i=1}^n [\hat{\mu}_i(q_1) - \hat{\mu}_i(q_0)], \quad \hat{\mu}_i(q) = \int y \hat{Q}_i^{y | a}(y | q) dy. \quad (3)$$

Here, the outer average is an explicit loop over units. The term $\hat{\mu}_i(q)$ is produced by the estimator attached to the conditional Q -node $Q_i^{y | a}$. For parametric families the integral is analytic or quadrature-based; for simulation-based components the backend draws Monte Carlo samples and averages the predicted outcomes. More complex graphs use the same evaluator, but the AST contains more density factors and more marginalization nodes. Scalable estimation enters at exactly this point. Each $\hat{\mu}_i$ can be fitted from the sub-units of unit i , each conditional density factor can be estimated from its own parent set, and each branch of a sampled marginalization can be evaluated independently before the AST combines the results. The computational structure therefore mirrors the statistical structure: many local estimators are assembled into one causal functional.

The canonical motifs used in the experiments serve as validation cases for this machinery. They verify that pyAgrum identification, HCM formula adaptation, AST evaluation, and numerical evaluation agree on structures whose behaviour is known in advance. In the confounder

motif, for example, a latent unit variable affects both treatment and outcome mechanisms. The collapsed graph contains Q_i^a and $Q_i^{y|a}$ as unit-level objects, and the intervention is evaluated by averaging the per-unit response:

$$\mathbb{E}[Y | \text{do}(Q^a = q_*)] = \mathbb{E}_i \left[\int y Q_i^{y|a}(y | q_*) dy \right]. \quad (4)$$

In the frontdoor/interference motif, an intermediate unit-level variable carries part of the causal pathway and the identified functional contains nested sums over the mediator distribution. In the instrumental motif, the expression contains a reweighting term analogous to inverse probability weighting:

$$\mathbb{E}[Y | \text{do}(A \sim q_*^a)] = \mathbb{E}_{Q^{a|z}, Q^z} \left[\frac{q_*^a(a)}{Q^{a|z}(a|z)} Y \right], \quad (5)$$

provided the positivity conditions are met. In the backend, the denominator $Q^{a|z}(a|z)$ is a fitted density node. If this estimator is missing or has no support at the sampled value, the calculation becomes unstable. This is why the implementation reports both the symbolic formula and factor-level diagnostics.

In practice, a symbolic formula may contain high-variance conditional density factors. This can happen with multi-parent Q -variables, such as $Q^{m|a,e,g,l}$, where the outcome distribution depends jointly on treatment, ethnicity, gender, and lunch status. The implementation records such cases and, in the normalized-factor runs studied below, rescales unstable multiplicative factors before evaluation. If the identified functional is $\mathfrak{F}$, the evaluated functional becomes a numerically stabilized object $\tilde{\mathfrak{F}}$ with the same graph structure but normalized factor weights. This is not a harmless implementation detail: normalizing $P(Z|W)$ changes the weighting induced by that factor and therefore changes the numerical functional being evaluated. We therefore report both the nominal identified formula and the factor normalization actually used in computation.

Experimental results

Canonical HCM motifs

The experimental validation starts with three canonical motifs: confounding, confounding with interference/frontdoor structure, and an instrumental-variable design. The data are generated from known HSCMs with $n = 100$ units and $m = 50$ sub-units, so that the estimated ATE can be compared with a known target rather than only with another estimator. Table 1 shows that the numerical evaluator recovers the intended causal contrast in these simple settings.

Table 1 ATE recovery on canonical HCM motifs.

Motif	True ATE	Estimated ATE	Absolute error
Confounder	2.00	1.99	0.01
Confounder and interference	1.50	1.52	0.02
Instrument	3.00	2.97	0.03

The confounder experiment also varies the number of units. With $m = 50$ sub-units per unit, the estimate converges toward the true ATE as n increases from 10 to 200, while pooled OLS remains biased because it mixes unit-level latent heterogeneity with the treatment effect. This behaviour is important for interpretation: the HCM estimator is recovering the unit-level intervention encoded by the graph, rather than only fitting an association in the pooled student table.

Do-calculus gallery

The symbolic part of the pipeline is then checked on a gallery of identifiable binary HCM configurations. Each configuration is identified automatically, converted to an AST, and evaluated numerically. Because the numerical ATE recovery for the three non-zero motifs is already reported in Table 1, the gallery is shown graphically rather than repeated as a second numerical table. Figures 1–3 display the corresponding complete graph transformations, from the original HCM to the collapsed and augmented or marginalized graphs used by do-calculus.

Nine benchmark scenarios

Beyond the canonical examples, the benchmark suite includes nine scenarios grouped into binary HCMs, spatio-temporal HCMs, and real-world-inspired synthetic examples. These experiments are designed not only to show that the estimator can succeed, but also to identify regimes where the empirical estimand differs from the theoretical target. Representative results are reported in Table 2.

The spatio-temporal experiments are particularly informative because the naive OLS estimate can be five to seven times larger than the true effect when a latent unit variable simultaneously affects baseline outcomes and treatment probability. When the empirical estimand matches the theoretical one, the per-unit HCM estimator removes this source of bias. When treatment interacts with context, however, the estimator reports the corresponding empirical target; this change in target is part of the interpretation rather than a numerical failure.

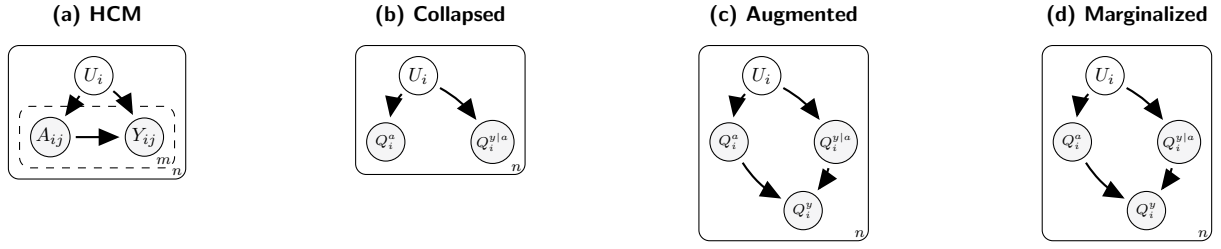


Figure 1 Complete graph transformations for the confounder motif. The collapsed graph exposes Q_i^a and $Q_i^{y|a}$ as unit-level random variables confounded by U_i ; augmentation adds the target Q_i^y .

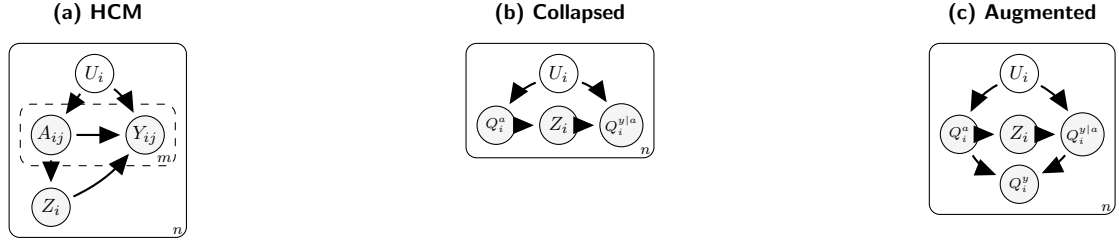


Figure 2 Complete graph transformations for the confounder-and-interference motif. The collapsed graph has a frontdoor-like path $Q_i^a \rightarrow Z_i \rightarrow Q_i^{y|a}$, and augmentation adds the target Q_i^y .

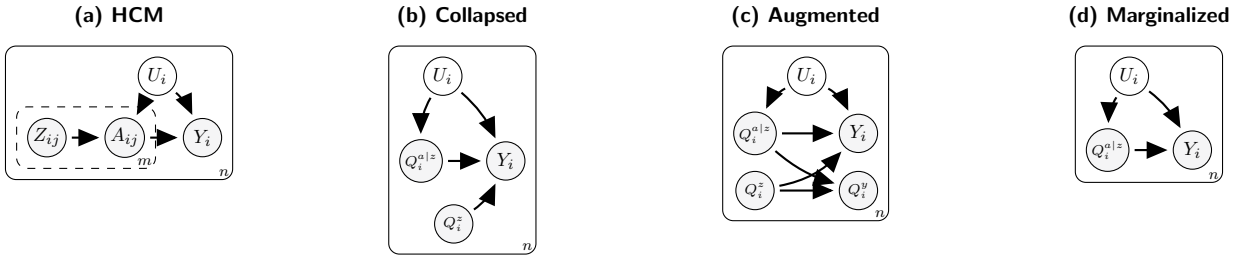


Figure 3 Complete graph transformations for the instrument motif. Marginalization removes Q_i^z , yielding the graph on which the do-calculus formula is evaluated.

Table 2 Representative benchmark scenarios.

Scenario	Ground truth	HCM/per-unit	Naive OLS
Binary confounder	0.315	0.363	—
Spatio-temporal confounder	5.00	5.10	27.1
Spatio-temporal dynamics	5.00	5.10	33.0
NLSY-style wages	0.35 log	0.34	0.70

Factor diagnostics

The experiments also expose a central operational issue: an identified formula may contain multi-parent Q -variables whose density factors are numerically unstable. In the normalized-factor runs, the evaluator rescales these multiplicative factors to keep the product stable and comparable across interventions. In the ExactBIC STAR run, the normalized factor was the gender-dependent term $P(Q^y | a, g | S)$. The diagnostic is scientifically useful only when it is read at the factor level: factor levels show which terms dominate the product numerically, while unit-context factor changes show which term changes between the two interventions.

Real data analysis: Project STAR

Data and hierarchical representation

Project STAR randomized kindergarten students within schools to small classes, regular classes, or regular classes with an aide. The full STAR-and-Beyond public-use dataset is documented by Achilles et al. (2008) and distributed through Harvard Dataverse at this link; it contains the raw student- and school-level records from the longitudinal Tennessee experiment, including demographics, class assignments, school identifiers, teacher information, and achievement outcomes. For the HCM analysis we use the kindergarten cohort from this full dataset

after removing missing values and balancing the data to 10 students per class, yielding 5,745 students, 322 classes, and 79 schools. The main outcome in this paper is Mathematics (M), measured by the STAR score `gktmathss`, but Reading (Y), measured by `gktreadss`, is retained because it enters some discovered graph structures and appears in the identified HCM formulas. Classes are treated as units and students as sub-units. The treatment is the student-level small-class assignment, written $A_{ij} = \mathbf{1}\{\text{small class}\}$, with student covariates given by Gender (G_{ij}), Ethnicity (E_{ij}), and Free-lunch status (L_{ij}). School urbanicity (S_i) is treated as an observed class-level covariate, while Unobserved class heterogeneity (U_i) represents latent class-level variation. The intervention is therefore not a row-level replacement of A_{ij} , but a class-level intervention $\text{do}(Q^a = q^a_*)$ that fixes the distribution of small-class assignment within the class.

Symbol	Level	Definition
A	student	Small-class assignment, <code>stark</code>
M	student	Mathematics score, <code>gktmathss</code>
Y	student	Reading score, <code>gktreadss</code>
G	student	Gender indicator, <code>gender</code>
E	student	Ethnicity indicator, <code>ethnicity</code>
L	student	Free-lunch status, <code>lunchk</code>
S	class	School urbanicity, <code>schoolk</code>
U	latent	Unobserved class heterogeneity

Table 3 STAR variables used in the HCM analysis.

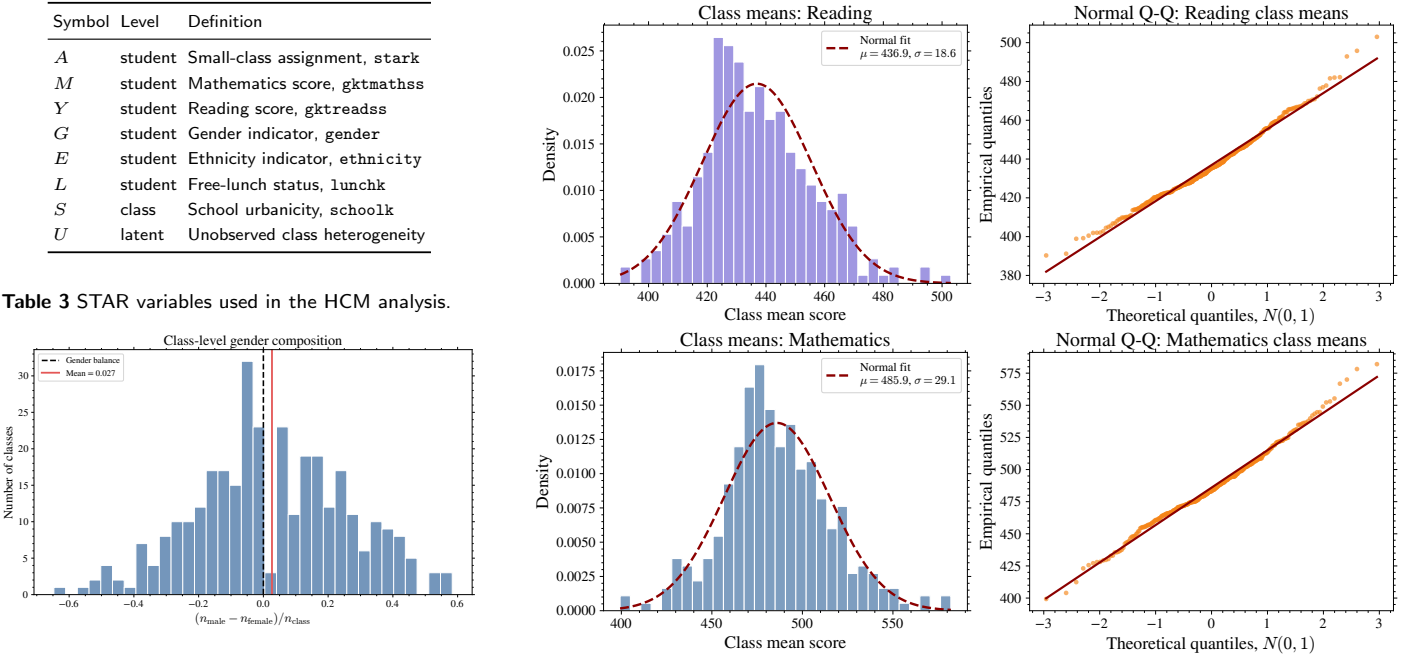


Figure 4 Class-level STAR diagnostics. Left: gender heterogeneity across classes, showing why an additive individual gender coefficient does not represent the full distribution of class composition. Right: reading and mathematics class-level score means, with histograms and Q-Q plots used to assess Gaussian plug-in choices for Q -variables.

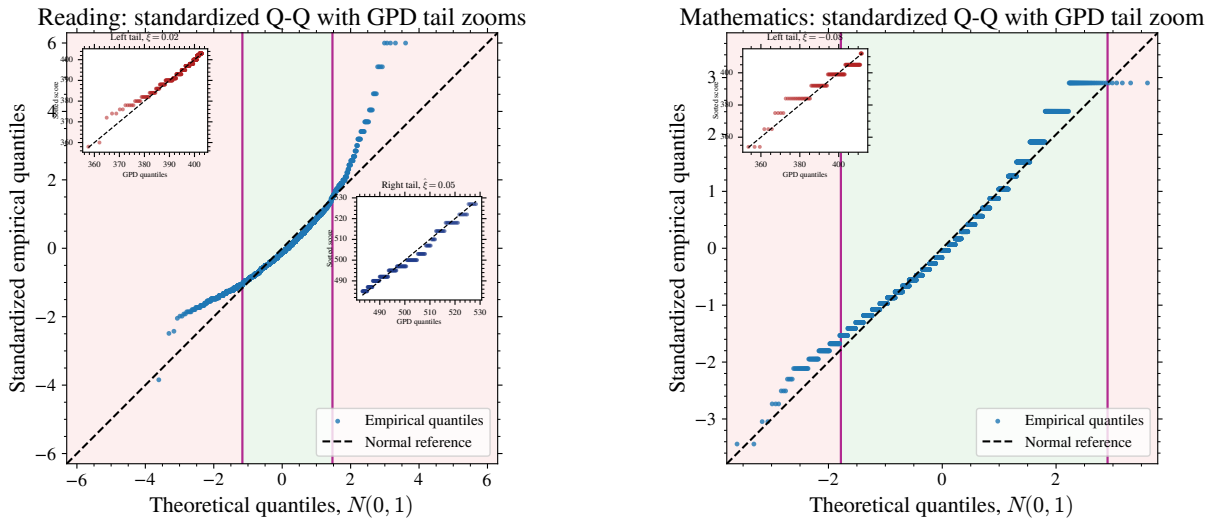


Figure 5 Heavy-tail Q-Q diagnostics for Reading (Y) and Mathematics (M). The standardized Q-Q plots and GPD tail zooms show departures from a global Gaussian approximation in the score extremes.

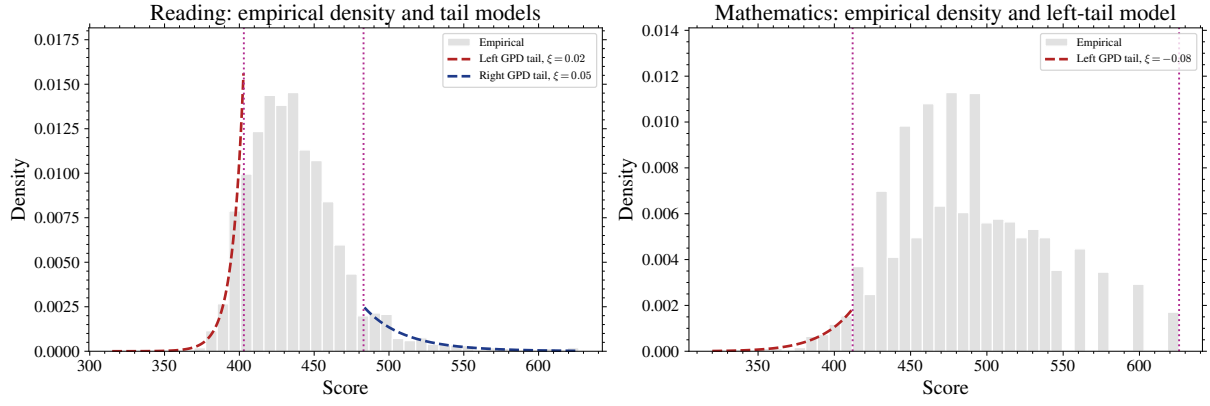


Figure 6 Heavy-tail histogram diagnostics for Reading (Y) and Mathematics (M). Empirical score densities are shown with GPD-type tail fits, confirming that the STAR score distributions have heavy non-Gaussian tails.

Baseline regressions

As baselines, we reproduce the standard student-level regression contrasts used in the STAR literature and compare them with a non-causal hierarchical reading of the data. The OLS replication starts from the pooled contrast

$$M_i = \alpha + \beta_1 \text{small}_i + \beta_2 \text{reg_aide}_i + \varepsilon_i, \quad (6)$$

then adds school fixed effects,

$$M_i = \alpha + \beta_1 \text{small}_i + \beta_2 \text{reg_aide}_i + \delta_{\text{school}(i)} + \varepsilon_i, \quad (7)$$

and finally includes student and teacher controls. The richest specification is

$$M_i = \alpha + \beta_1 \text{small}_i + \beta_2 \text{reg_aide}_i + \gamma^\top X_i + \delta_{\text{school}(i)} + \varepsilon_i, \quad (8)$$

where X_i includes race, gender, free-lunch status, teacher race, teacher experience, and teacher education. The coefficient β_1 is therefore a student-level small-class contrast, conditional on the chosen controls.

Table 4 OLS estimates of the small-class effect on kindergarten mathematics.

Model	$\hat{\beta}_{\text{small}}$	SE	p -value	R^2
No controls	8.096	1.597	4.0×10^{-7}	0.006
School fixed effects	9.499	1.463	8.5×10^{-11}	0.219
Student controls	9.380	1.415	3.4×10^{-11}	0.268
Full OLS replication	9.432	1.419	3.0×10^{-11}	0.269

The instrumental-variable replication uses assignment indicators as instruments for actual class size. In first stage,

$$\text{classsize}_i = \pi_0 + \pi_1 \text{small}_i + \pi_2 \text{reg_aide}_i + \Pi^\top X_i + \delta_{\text{school}(i)} + \nu_i, \quad (9)$$

and the second stage estimates

$$M_i = \alpha + \theta \widehat{\text{classsize}}_i + \gamma^\top X_i + \delta_{\text{school}(i)} + \varepsilon_i, \quad (10)$$

This gives a second-stage coefficient of -1.219 mathematics points per additional student (SE 0.163). The corresponding non-causal hierarchical regression baseline can be written as a random-intercept or Bayesian OLS model,

$$M_{ij} = \alpha + \beta_1 \text{small}_{ij} + \beta_2 \text{reg_aide}_{ij} + \gamma^\top X_{ij} + b_{\text{class}(i)} + \varepsilon_{ij}, \quad b_c \sim \mathcal{N}(0, \sigma_b^2). \quad (11)$$

This model is valuable because it acknowledges between-class heterogeneity, but it treats that heterogeneity as a random effect rather than as a distributional causal mechanism. OLS, IV, and hierarchical Bayesian/OLS baselines therefore include gender and class variation descriptively, but they do not treat class composition or treatment-by-gender mechanisms as intervention-level distributions. This distinction motivates the HCM diagnostics below.

HCM under discovered graphs

ExactBIC: nominal and effective graph

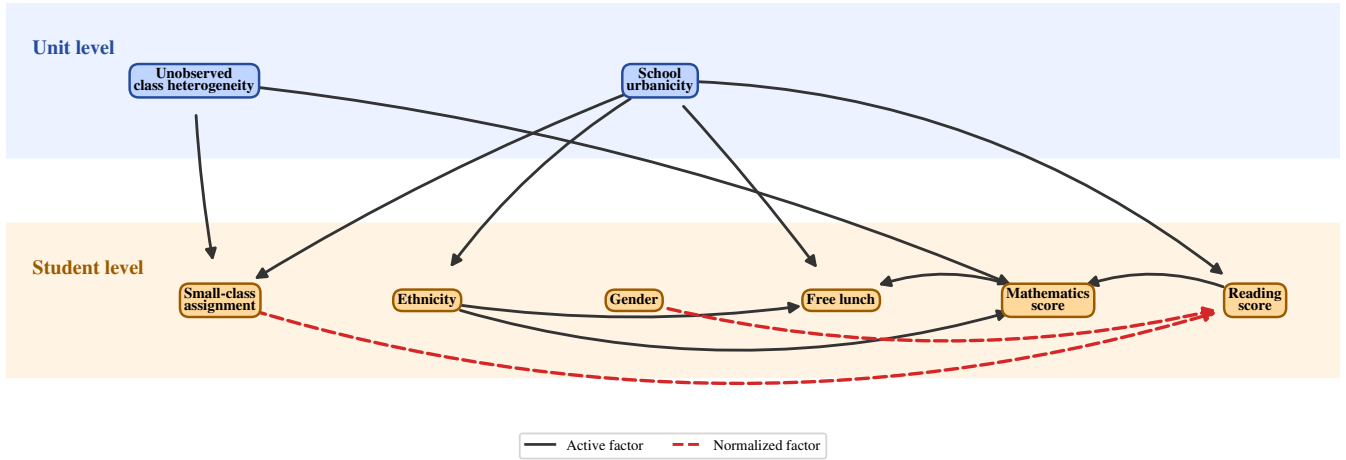


Figure 7 ExactBIC-derived HCM graph with normalized factors. Dashed red arrows mark dependencies whose multiplicative factors are normalized for numerical stability in the stabilized evaluation.

ExactBIC after HCM collapse/augmentation: Q-variable graph

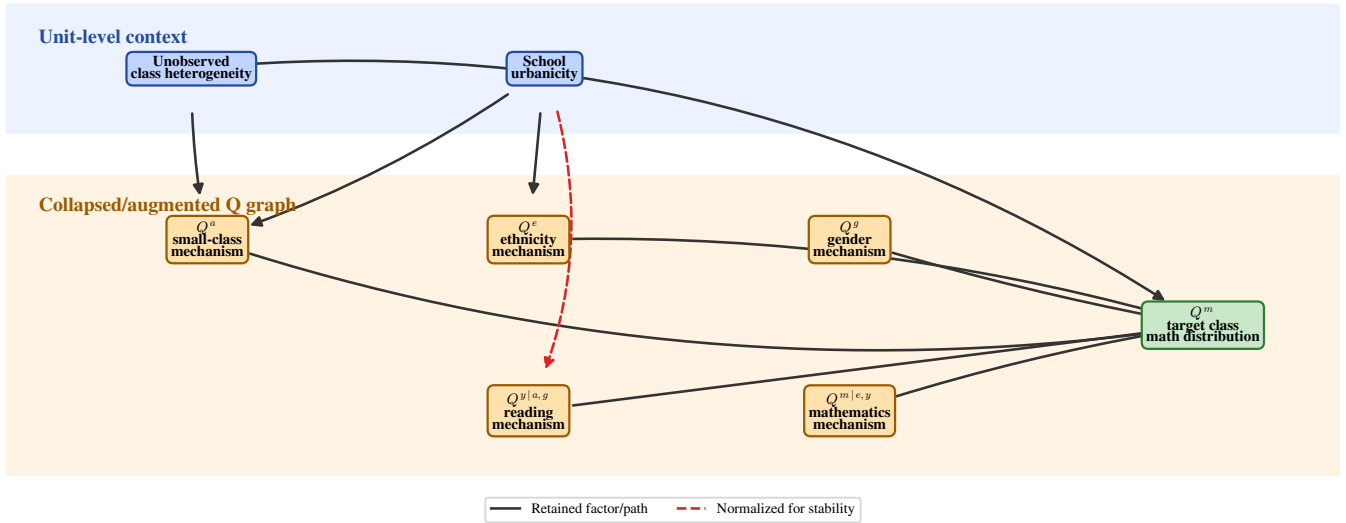


Figure 8 ExactBIC graph after HCM collapse and augmentation for the mathematics target. Student-level variables are represented by class-level mechanisms Q ; the green node is the target class-level Mathematics distribution Q^m evaluated under $\text{do}(Q^a = q_\star^a)$. The dashed red arrow indicates the $P(Q^y | a, g | S)$ factor normalized in the stabilized ExactBIC run.

The STAR HCM analysis starts from graph structures learned on the flat discovery variables

(readk, mathk, stark, gender, ethnicity, lunchk, schoolk).

For the ExactBIC run, we used the score-based DAG stored in the discovery output, with the school-urbanicity/ethnicity relation oriented as Urbanicity (S) $\rightarrow$ Ethnicity (E) for substantive interpretability. The directed edges were translated through the fixed map

stark $\mapsto$ Small-class assignment (A), readk $\mapsto$ Reading (Y), mathk $\mapsto$ Mathematics (M),

gender $\mapsto$ Gender (G), ethnicity $\mapsto$ Ethnicity (E), lunchk $\mapsto$ Free lunch (L), schoolk $\mapsto$ Urbanicity (S).

Thus the ExactBIC graph used here contains the translated dependencies Reading (Y) $\rightarrow$ Mathematics (M), Mathematics (M) $\rightarrow$ Free lunch (L), Small-class assignment (A) $\rightarrow$ Reading (Y), Gender (G) $\rightarrow$ Reading (Y), Ethnicity (E) $\rightarrow$ Mathematics (M), Ethnicity (E) $\rightarrow$ Free lunch (L), Urbanicity (S) $\rightarrow$ Ethnicity (E), Urbanicity (S) $\rightarrow$ Reading (Y), Urbanicity (S) $\rightarrow$ Small-class assignment (A), and Urbanicity (S) $\rightarrow$ Free lunch (L). Before identification, the HCM construction also adds the latent class-level confounding edges Unobserved class heterogeneity (U) $\rightarrow$ Small-class assignment (A) and Unobserved class heterogeneity (U) $\rightarrow$ Mathematics (M), because the target outcome in this paper is Mathematics (M). The resulting hierarchical graph is then collapsed and augmented for Q^m (the class-level Mathematics mechanism), giving the Q -variable graph in Figure 8; this transformed graph is passed to pyAgrum and evaluated with the plug-in families used in the run: Bernoulli models for Small-class assignment (A), Gender (G), Ethnicity (E), and Free lunch (L), a categorical model for Urbanicity (S), and two-component Gaussian mixtures for Reading (Y) and Mathematics (M).

After this translation, the class-as-unit HCM v2 model gives the estimates reported in Table 5.

Table 5 HCM estimates for STAR mathematics outcome.

Run	$E[\text{do}(1)]$	$E[\text{do}(0)]$	ATE
ExactBIC	2406.65	2366.69	39.96
ExactBIC (normalized factors)	1788.10	1766.47	21.63

These values should not be read as direct replacements for the student-level OLS coefficient. The OLS coefficient is a student-level contrast for a binary treatment indicator, conditional on observed covariates and school fixed effects. The HCM estimand is instead a class-level intervention on Q^a , with additional dependence through class-specific distributions. The HCM magnitudes therefore describe the behaviour of the translated hierarchical estimand, and the difference between the raw and normalized-factor rows shows that the result is sensitive to the scaling of multiplicative factors retained in the identified formula.

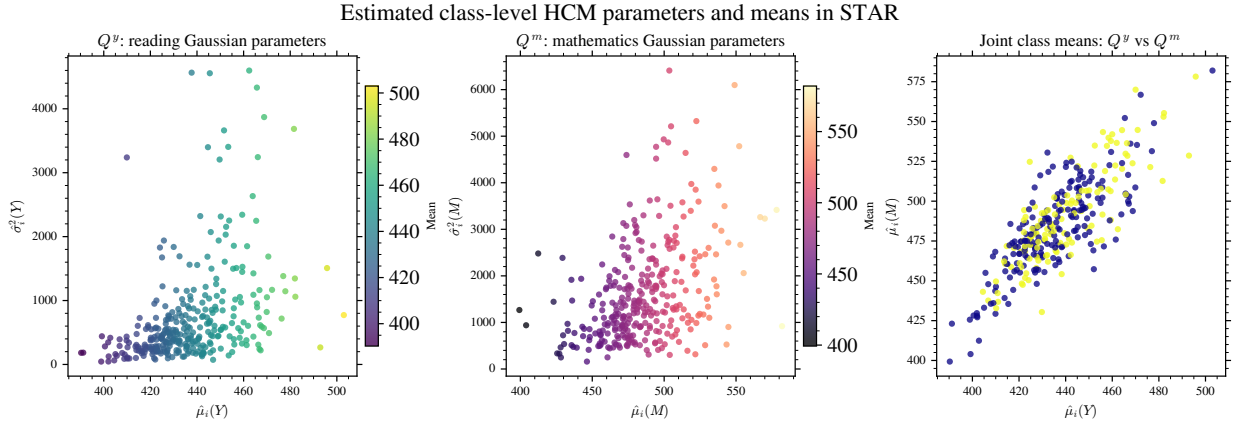


Figure 9 Estimated class-level HCM response parameters and means in STAR. Reading (Q^y) and Mathematics (Q^m) are represented by Gaussian summaries (μ_i, σ_i^2), and the right panel reports the joint class means used to inspect the relationship between the two score mechanisms.

Identified formulas for STAR

The ExactBIC-derived graph gives the following functional after the translated graph has been collapsed and augmented for the mathematics outcome:

$$\begin{aligned} \tau_{\text{BIC}}(q^a) = & \sum_{q^e, q^g, q^m | e, y, q^y | a, g, s} p(q^e | s) p(q^y | a, g | s) p(q^m | e, y) \\ & \times \mathbb{E} \left[Q^m | Q^a = q^a, Q^e, Q^g, Q^m | e, y, Q^y | a, g \right] p(q^g) p(s). \end{aligned} \quad (12)$$

In this expression, $p(q^m | e, y)$ is the class-level Mathematics (M) distribution induced by the discovered Reading (Y) $\rightarrow$ Mathematics (M) and Ethnicity (E) $\rightarrow$ Mathematics (M) parents, $p(q^e | s)$ encodes the substantively oriented Urbanicity (S) $\rightarrow$ Ethnicity (E) relation, and $p(q^y | a, g | s)$ comes from the Reading (Y) mechanism with Small-class assignment (A), Gender (G), and Urbanicity (S) in the translated graph. The expectation term is evaluated twice, at $Q^a = 1$ and $Q^a = 0$, and the remaining Q -variables are marginalized by the product of fitted density factors. The formula therefore makes clear why multi-parent Q -estimation matters: the identified effect is a product of conditional densities and outcome-response factors, not a single regression coefficient.

Effective graphs and factor normalization

The normalized-factor runs make explicit the difference between a nominal identified functional and the stabilized functional actually evaluated by the numerical code. For ExactBIC, the normalized term was $P(Q^{y|a,g} | S)$, which is the distribution of the class-level Reading (Y) mechanism conditional on Small-class assignment (A) and Gender (G), further conditioned by Urbanicity (S). Normalizing such a factor does not remove variables from the nominal causal graph, but it rescales the corresponding weighting in the product that is numerically evaluated. The effective graph in Figure 7 visualizes this distinction: dashed red arrows are dependencies present in the nominal graph whose factor contribution is normalized in the stabilized functional.

The ExactBIC gender factor analysis clarifies what this means numerically. The normalized factor $P(Q^{y|a,g} | S)$ is stabilized before the two intervention evaluations, so its raw scale does not dominate the product. It nevertheless marks an important part of the estimand: the graph says that the class-level Reading (Y) response should depend on Small-class assignment (A), Gender (G) composition, and Urbanicity (S). When the same run is analysed in unit context, the only non-zero factor change comes from the fitted response term $P(Q^y | Q^a, Q^g, Q^{y|a,g})$, namely Reading (Y) conditional on Small-class assignment (A), Gender (G), and the class-level Reading mechanism $Q^{y|a,g}$, whose mean changes from 434.83 under $\text{do}(Q^a = 0)$ to 440.15 under $\text{do}(Q^a = 1)$. The resulting delta, 5.32, explains why the gender mechanism matters even when the conditional factor is normalized for stability.

Diagnostics and interpretation

The STAR analysis also illustrates why diagnostics are essential. The score distributions have heavy, non-Gaussian tails, so Figures 5 and 6 report tail diagnostics rather than relying only on a global Gaussian approximation. This matters for the plug-in analysis because a Gaussian model can describe the central mass of Reading (Y) and Mathematics (M) while still underrepresenting the probability of extreme classroom scores. The gender-composition diagnostic in Figure 4 shows that class composition varies substantially across classes; an OLS coefficient on **female** adjusts for an average individual difference, but it does not represent the distribution of class composition. HCM explicitly represents such within-class distributions through the estimated probability Q -quantities in Figure 10 and the response parameters and means in Figure 9. The ExactBIC factor analysis above shows how gender composition enters the intervention path through $Q^{y|a,g}$. The current implementation remains two-level and treats schools only through unit-level covariates rather than as a full student–class–school hierarchy. This is the main substantive lesson of the STAR analysis. Earlier non-causal hierarchical or graphical summaries could describe that classes differ, but they did not make class heterogeneity part of an identified intervention functional. By combining class-level Q -heterogeneity with factor analysis of the identified formula, the HCM pipeline shows that part of the small-class effect can be hidden when class composition and multi-parent distributional factors are treated only as background variation. In that sense, the previous hierarchical graph analysis tended to smooth or understate the intervention contrast, whereas the causal HCM analysis gives a new interpretation of STAR: the effect of class size is not only an average student-level coefficient, but a distributional class-level effect whose magnitude depends on which class mechanisms are included, estimated, or normalized in the plug-in functional.

The substantive conclusion is therefore conservative. HCM does not simply “improve” OLS by producing a larger number; it changes the estimand from a student-level coefficient to a distributional class-level intervention. In STAR this change is scientifically meaningful, but it also makes graph choice, positivity, and plug-in completeness part of the reported result rather than secondary implementation details.

Conclusion

We developed a scalable automatic identification and estimation implementation for hierarchical causal inference. The pipeline starts from a hierarchical graph, applies the HCM transformations needed for identification, obtains a symbolic do-calculus estimand from pyAgrum, adapts it into closed-form HCM formulas, and evaluates the associated AST using plug-in density estimators. This turns HCM from a formal identification framework into an operational workflow for nested data. The key computational point is that the AST does not leave the formula as an opaque expression: it decomposes the estimand into many local conditional densities, response models, and marginalization branches that can be estimated independently and then recombined into a single causal functional. The implementation is available at [AI-vidence/hierarchicalcausalmodels](#), so the graph transformations, identification calls, estimators, diagnostics, and STAR replication artefacts can be inspected and reused.

The experiments validate the pipeline on known structures and reveal the practical regimes in which estimation is stable. The STAR application demonstrates the methodological payoff: when the intervention is assigned at the class level and outcomes are measured at the student level, the natural estimand is hierarchical. OLS, IV, and non-causal hierarchical summaries remain essential benchmarks, but they do not encode interventions on within-class treatment distributions or class-composition mechanisms. The same framework also clarifies why large hierarchical applications are computationally feasible: adding units, factors, or sampled branches increases the number of local estimation tasks, but it does not require redefining the estimand or replacing it by a flat pooled regression.

The remaining limitations point directly to the next statistical developments. The present implementation is two-level, whereas STAR is naturally a student–class–school system; extending scalable HCM estimation to deeper hierarchies is therefore necessary for a fully faithful education application. Multi-parent Q -density estimation is the main statistical bottleneck, because missing or weakly estimated factors change the effective estimand rather than merely adding numerical noise. Uncertainty quantification for normalized or otherwise stabilized functionals also remains open. Future work should therefore develop robust multi-parent estimators, extend the implementation to deeper hierarchies, preserve the parallel structure of the AST evaluator, and connect the numerical backend to symbolic DSLs such as `y0` so that formula manipulation and estimation can share a common representation.

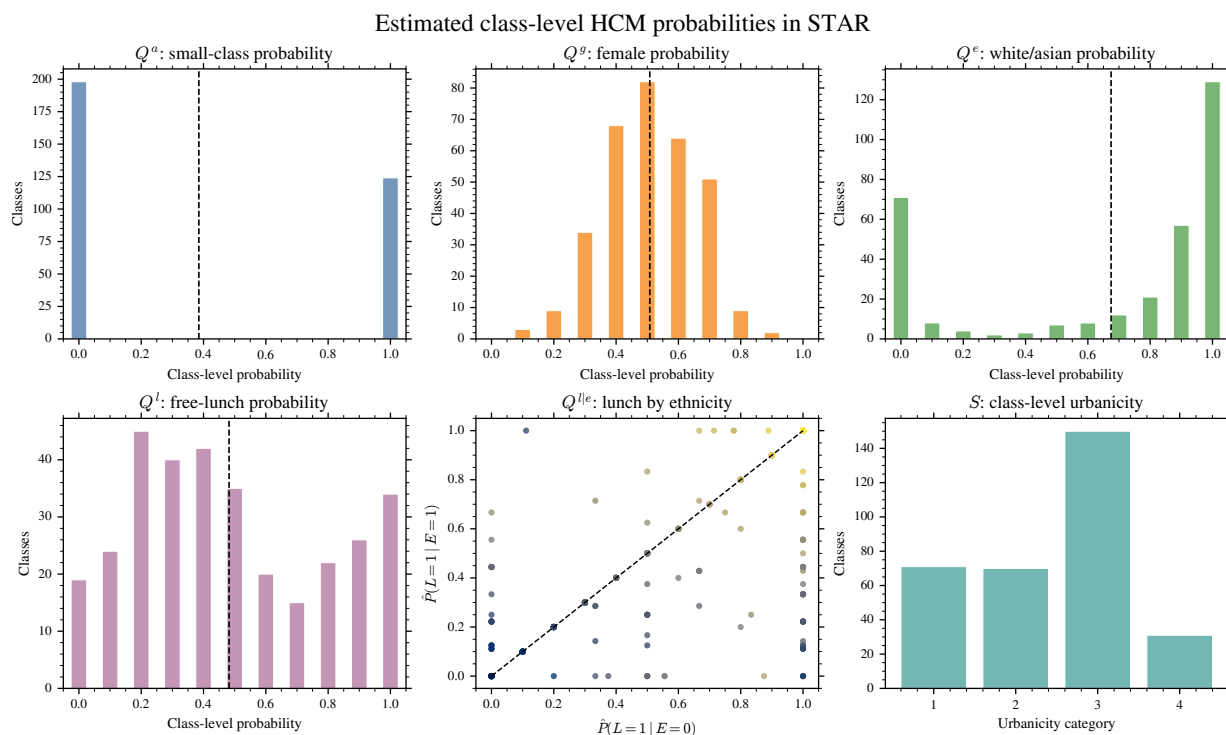


Figure 10 Estimated class-level HCM probability quantities used in the STAR plug-in analysis. Binary sub-unit variables are represented by class-level Bernoulli probabilities for Small-class assignment (Q^a), Gender (Q^g), Ethnicity (Q^e), and Free lunch (Q^l); $Q^{l|e}$ summarizes the Free-lunch mechanism conditional on Ethnicity; and S is the observed class-level Urbanicity variable.

Author contributions

All authors contributed to methodology, software, experiments, and writing.

Data availability

The full Project STAR public-use data are available as the STAR-and-Beyond dataset documented by Achilles et al. (2008) and hosted by Harvard Dataverse at this link. Code and generated artefacts are maintained at this link.

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Software notes

The implementation provides graph transformations, do-calculus identification, AST evaluation, parametric and neural estimators, and STAR replication scripts. The current draft reports the stable artefacts used for the STAR analysis. The code is available at [this link](#).